Macroeconomics I — MPE Insper

Prova 01 — detailed solution

Graded answer: Q1 1.0/1.5, Q2 3.1/5.5, Q3 0.5/3.0

How to read this. Each question and item opens with its statement reproduced verbatim from the exam, in Portuguese as printed. Each question then has a Notation list. Every equation a later step uses carries a number, and each step says which equations it uses. Q2 and Q3 keep the exam's own equation labels. Each item ends with the intuition, what the grader penalised in the submitted answer, and what earns the points. The item-by-item diagnosis is in `prova-01-error-analysis.md`.

Question 1 — National accounts (1.5 pt)

Statement, as printed in the exam:

> **1. (1,5 pontos)** Considere uma economia simplificada, sem impostos, gastos do governo e sem comércio exterior. Nela há dois tipos de firmas: a que produz insumos necessários à produção dos bens, à cargo da outra firma, vendidos às famílias. Suponha que num determinado ano, a firma produtora de insumos (Firma A) produziu \$600 que foram vendidos à Firma B, que produziu bens vendidos às famílias no valor de \$1.000.

Notation.

- A, B, N: the input firm, the final-goods firm and (from 1.b) the social network.
- VA_j: value added of firm j, its gross output minus the intermediate inputs it uses up.
- C, I, G, NX: household consumption, investment, government purchases and net exports (final uses). Here $G = NX = 0$.
- Factor income: wages paid to workers plus profits (operating surplus) left to owners.

Step — 0, the three definitions used below. GDP is the market value of the *final* goods and services produced in the period. It can be computed three ways, which must agree:

$$\text{GDP} = \textstyle\sum_j VA_j, \tag{G1}$$

$$VA_j = \text{gross output}_j - \text{intermediate inputs}_j, \tag{G2}$$

$$\text{GDP} = \textstyle\sum_j (\text{wages}_j + \text{profits}_j), \tag{G3}$$

$$\text{GDP} = C + I + G + NX. \tag{G4}$$

A good is *intermediate* when it is used up in producing another good within the period.

Statement, as printed in the exam:

> **1.a (0,5 ponto)** Calcule o PIB desta economia, explicando precisamente o cálculo e o conceito utilizado.

Step — 1, classify. The \$600 of inputs is used up by B within the year, so it is intermediate.

Step — 2, product approach. Using (G2) for each firm (A buys no inputs):

$$VA_A = 600 - 0 = 600,$$
$$VA_B = 1000 - 600 = 400.$$

Summing with (G1):

$$\text{GDP} = 600 + 400 = 1000.$$

Step — 3, income approach. Each firm's value added is paid out to its workers and owners. A's factors earn 600 and B's earn 400, so by (G3):

$$\text{GDP} = 600 + 400 = 1000.$$

Step — 4, expenditure approach. The only final use is household consumption, $C = 1000$, with $I = G = NX = 0$. By (G4):

$$\text{GDP} = 1000 + 0 + 0 + 0 = 1000.$$

Economic intuition. Adding $600 + 1000 = 1600$ would count the inputs twice: once when A sells them and again inside the price of B's goods. Netting out intermediate inputs in (G2) is what makes the three approaches agree.

Ref: Kurlat (2020), §1.1–1.2, pp. 13–19 (the three approaches; Example 1.6, p. 19).

What earns the points.

- 1000 with the final-good or value-added concept stated (full marks were given).

Statement, as printed in the exam:

> **1.b (0,5 ponto)** Considere agora que nesta economia a Firma B contrata o serviço de uma rede social recentemente criada que é gratuita para seus usuários e financiada por publicidade. A Firma B gasta \$100 em publicidade na rede social mas isto não gera ainda impacto nas suas vendas no período, que fica no patamar inicial. Calcule, justificando, o PIB desta economia na ótica do produto, da renda e do dispêndio, comparando-o com o valor obtido no item anterior (1.a), explicando o que esta comparação indica.

Step — 1, classify the advertising. Advertising is a service bought *by a firm* and used up in its production in the period, so it is an intermediate input of B. The service N gives its users has a price of zero; no transaction takes place, so it is not in GDP.

Step — 2, product approach. Using (G2), with B now using two inputs:

$$\begin{aligned} VA_A &= 600 - 0 = 600, \\ VA_B &= 1000 - 600 - 100 = 300, \\ VA_N &= 100 - 0 = 100. \end{aligned}$$

Summing with (G1):

$$\text{GDP} = 600 + 300 + 100 = 1000.$$

Step — 3, income approach. A's factors earn 600. B's profit falls by the 100 it pays N, so B's factors earn 300. N's workers and owners earn 100. By (G3):

$$\text{GDP} = 600 + 300 + 100 = 1000.$$

Step — 4, expenditure approach. Final uses are unchanged ($C = 1000$); the advertising is not a final use. By (G4):

$$\text{GDP} = 1000.$$

Approach	A	B	N	GDP
Product, (G1)–(G2)	600	300	100	1000
Income, (G3)	600	300	100	1000
Expenditure, (G4)		$C = 1000$		1000

Step — 5, comparison. GDP is the same as in 1.a. Only the *distribution* of income moved: 100 went from B's factors to N's. Users do get value from the free service, but GDP does not record it, because it has no market price. This is a limit of GDP as a welfare measure.

Economic intuition. A new firm appears and GDP does not rise, because its whole output is an input into a product whose final value did not change.

Where the marks went. The answer gave 1000 with one sentence. The item asked for all three approaches; the grader wrote "3 óticas??".

Ref: Kurlat (2020), §1.2 (intermediate vs final); ch. 2 (what GDP leaves out).

What earns the points.

- Advertising classified as intermediate.
- All three approaches computed and shown to agree.
- The comparison: same GDP, redistribution, unmeasured value of the free service.

Statement, as printed in the exam:

> **1.c (0,5 ponto)** Suponha agora que a rede social utilize integralmente a receita de $100 para produzir um novo software (e nada mais se altere em relação ao que foi descrito em 1.b). Calcule, justificando, o PIB desta economia na ótica do produto, da renda e do dispêndio, comparando-o com o valor obtido no item anterior (1.b), explicando o que esta comparação indica.

Step — 1, classify the software. N pays $100 to programmers who write software for N's own use. The software is a durable productive asset, used for more than one period. National accounts record it as *gross fixed capital formation* (investment in intellectual property), valued at its cost of 100. It is a final good.

Step — 2, product approach. N's gross output is now ads plus software, $100 + 100 = 200$, with no intermediate inputs. Using (G2):

$$VA_N = 200 - 0 = 200,$$
$$VA_A = 600, \qquad VA_B = 300 \text{ (as in 1.b)}.$$

Summing with (G1):

$$\text{GDP} = 600 + 300 + 200 = 1100.$$

Step — 3, income approach. N pays 100 in wages. Its operating surplus is its output minus wages:

$$\text{profit}_N = 200 - 100 = 100.$$

So N's factor income is $100 + 100 = 200$. By (G3):

$$\text{GDP} = 600 + 300 + 200 = 1100.$$

Step — 4, expenditure approach. Nothing else changes, so $C = 1000$. Investment is now the software, $I = 100$. By (G4):

$$\text{GDP} = 1000 + 100 = 1100.$$

Approach	A	B	N	GDP
Product, (G1)–(G2)	600	300	200	1100
Income, (G3)	600	300	wages 100 + profit 100	1100
Expenditure, (G4)		$C = 1000$, $I = 100$		1100

Step — 5, comparison. GDP rises by 100 relative to 1.b because a new *final* good, an investment good, was produced. If the software were expensed as an intermediate cost, GDP would stay at 1000. Capitalising software is the modern national-accounts convention, and it is what makes the three approaches agree at 1100.

Economic intuition. In 1.b the $100 was used up inside B's product. Here it is spent producing something that lasts, and producing a lasting asset adds to output.

Where the marks went. The answer reached 1100 but said the $100 becomes "wages and consumption". On the income side it is wages; on the expenditure side it is *investment*, since consumption is still 1000 by assumption. The three approaches were again missing.

Ref: Kurlat (2020), §1.2 (investment as a final use, pp. 17–19).

What earns the points.

- Software identified as investment, not consumption or intermediate.
- All three approaches with 1100.
- The comparison: the +100 comes from I.

Question 2 — Solow, golden rule and a two-period competitive equilibrium (5.5 pt)

Statement, as printed in the exam:

> **2. (5,5 pontos)** Considere uma economia descrita pelo modelo de Solow, onde as variáveis agregadas satisfazem as equações: (1) $Y_t = A_t \,.\, (K_t^{\sigma}) \,.\, (L_t^{1-\sigma})$, (2) $K_{t+1} = (1-\delta)K_t + s \,.\, Y_t$, (3) $L_{t+1} = (1+n)L_t$, (4) $A_{t+1} = A_t = A > 1$, sendo que $\{\sigma, \delta, s, n\} \in (0,1)$.

The statement's equations, numbered as in the exam, for citation below:

$$Y_t = A_t \, K_t^{\sigma} L_t^{1-\sigma}, \tag{1}$$
$$K_{t+1} = (1-\delta)K_t + s\, Y_t, \tag{2}$$
$$L_{t+1} = (1+n)L_t, \tag{3}$$
$$A_{t+1} = A_t = A > 1. \tag{4}$$

Notation.

- Y_t: output (GDP) in period t. K_t: capital stock. L_t: number of workers.
- A: total factor productivity, constant and > 1 by (4).
- $\sigma \in (0,1)$: elasticity of output with respect to capital, which under competition is capital's share of income.
- $\delta \in (0,1)$: depreciation rate. $s \in (0,1)$: saving rate (share of output saved). $n \in (0,1)$: growth rate of the labour force.
- $y_t \equiv Y_t/L_t$, $k_t \equiv K_t/L_t$, c_t: output, capital and consumption per worker.
- k_{SS}: steady-state capital per worker. s_{GR}: golden-rule saving rate.

Statement, as printed in the exam:

> **2.a (1 ponto)** Defina as variáveis estacionárias por letras minúsculas e encontre uma expressão para (k_{t+1}/k_t) e, então, o valor de estado estacionário de k, denotando-o por k_{SS}.

Step — 1, define the stationary variables. A is constant and L_t grows at rate n, so the variables that settle down are the per-worker ones:

$$y_t \equiv \frac{Y_t}{L_t}, \qquad k_t \equiv \frac{K_t}{L_t}. \tag{5}$$

Step — 2, intensive production function. Divide (1) by L_t, using $A_t = A$ from (4):

$$\begin{aligned} y_t &= \frac{A K_t^{\sigma} L_t^{1-\sigma}}{L_t} \\ &= A K_t^{\sigma} L_t^{-\sigma} \\ &= A \left(\frac{K_t}{L_t}\right)^{\sigma} \\ &= A k_t^{\sigma}. \end{aligned} \tag{6}$$

Step — 3, accumulation per worker. Divide (2) by L_t:

$$\frac{K_{t+1}}{L_t} = (1-\delta)\frac{K_t}{L_t} + s\,\frac{Y_t}{L_t}.$$

Replace K_t/L_t by k_t from (5) and Y_t/L_t by $A k_t^{\sigma}$ from (6):

$$\begin{aligned} \frac{K_{t+1}}{L_t} &= (1-\delta)\,(k_t) + s\,(A k_t^{\sigma}) \\ &= (1-\delta)k_t + s\, A k_t^{\sigma}. \end{aligned} \tag{7}$$

The left side has K_{t+1} over last period's labour. Multiply and divide by L_{t+1}, then use (3):

$$\begin{aligned}\frac{K_{t+1}}{L_t} &= \frac{K_{t+1}}{L_{t+1}} \cdot \frac{L_{t+1}}{L_t} \\ &= (1+n)\, k_{t+1}. \end{aligned} \tag{8}$$

Substitute (8) into the left side of (7):

$$\begin{aligned}\big((1+n)\, k_{t+1}\big) &= (1-\delta)k_t + s\, Ak_t^{\sigma} \\ (1+n)\, k_{t+1} &= (1-\delta)k_t + s\, Ak_t^{\sigma}. \end{aligned} \tag{9}$$

Step — 4, the ratio. Divide (9) by $(1+n)k_t$:

$$\frac{k_{t+1}}{k_t} = \frac{1-\delta+s\, Ak_t^{\sigma-1}}{1+n}. \tag{10}$$

Step — 5, steady state. In a steady state $k_{t+1} = k_t = k$, so the left side of (10) is (1):

$$(1) = \frac{1-\delta+s\, Ak^{\sigma-1}}{1+n}.$$

Multiply by $1+n$ and solve for k:

$$\begin{aligned}1+n &= 1-\delta+s\, Ak^{\sigma-1} \\ n+\delta &= s\, Ak^{\sigma-1} \\ k^{1-\sigma} &= \frac{sA}{n+\delta}. \end{aligned}$$

Raise both sides to the power $1/(1-\sigma)$:

$$\boxed{k_{SS} = \left(\frac{sA}{n+\delta}\right)^{\frac{1}{1-\sigma}}} \tag{11}$$

Step — 6, stability. In (10), $\sigma < 1$ makes $k_t^{\sigma-1}$ fall as k_t rises. So the ratio is above 1 when $k_t < k_{SS}$ and below 1 when $k_t > k_{SS}$: from any $k_0 > 0$, k_t converges to k_{SS}.

Economic intuition. In the steady state, saving per worker, sAk^{σ}, exactly covers the capital needed to equip new workers (nk) and to replace worn-out capital (δk).

What earns the points.

- (6) and the $(1+n)$ factor in (8) derived, not assumed.
- The ratio (10) and k_{SS} (11) (full marks were given).

Statement, as printed in the exam:

> **2.b (1 ponto)** Demonstre qual é a taxa de poupança maximiza o consumo de estado estacionário (a taxa da chamada regra de ouro, s_{GR}).

Step — 1, consumption per worker. Consumption is the part of output not saved:

$$c_t = (1-s)\, y_t. \tag{12}$$

Step — 2, steady-state output. Evaluate (6) at k_{SS} from (11):

$$\begin{aligned}y_{SS} &= A\, k_{SS}^{\sigma} \\ &= A\left[\left(\frac{sA}{n+\delta}\right)^{\frac{1}{1-\sigma}}\right]^{\sigma}\end{aligned}$$

$$= A\left(\frac{sA}{n+\delta}\right)^{\frac{\sigma}{1-\sigma}}. \tag{13}$$

The exponent is $\frac{\sigma}{1-\sigma}$: output depends on k_{SS} raised to σ.

Step — 3, steady-state consumption. Substitute (13) for y in (12):

$$\begin{aligned} c_{SS}(s) &= (1-s)\left(A\left(\frac{sA}{n+\delta}\right)^{\frac{\sigma}{1-\sigma}}\right) \\ &= (1-s)\,A\left(\frac{sA}{n+\delta}\right)^{\frac{\sigma}{1-\sigma}}. \end{aligned} \tag{14}$$

Step — 4, take logs. Take the log of (14):

$$\ln c_{SS} = \ln A + \ln(1-s) + \frac{\sigma}{1-\sigma}\Big[\ln s + \ln A - \ln(n+\delta)\Big]. \tag{15}$$

Only $\ln(1-s)$ and $\frac{\sigma}{1-\sigma}\ln s$ depend on s.

Step — 5, first-order condition. Differentiate (15) with respect to s and set it to zero:

$$-\frac{1}{1-s} + \frac{\sigma}{(1-\sigma)\,s} = 0. \tag{16}$$

Multiply (16) by $(1-\sigma)s(1-s) > 0$:

$$\begin{aligned} -(1-\sigma)s + \sigma(1-s) &= 0 \\ -s + \sigma s + \sigma - \sigma s &= 0 \\ \sigma - s &= 0. \end{aligned}$$

Hence

$$\boxed{s_{GR} = \sigma} \tag{17}$$

Step — 6, second-order condition. Differentiate (16) once more:

$$\frac{d^2 \ln c_{SS}}{ds^2} = -\frac{1}{(1-s)^2} - \frac{\sigma}{(1-\sigma)s^2} < 0.$$

So (15) is strictly concave in s, and (17) is the maximum.

Step — 7, cross-check through k. In a steady state, (9) with $k_{t+1} = k_t = k$ gives

$$sAk^{\sigma} = (n+\delta)\,k. \tag{18}$$

Substitute (6) for y in (12), then replace sAk^{σ} using (18):

$$\begin{aligned} c_{SS} &= (1-s)\big(Ak^{\sigma}\big) \\ &= Ak^{\sigma} - sAk^{\sigma} \\ &= Ak^{\sigma} - \big((n+\delta)k\big) \\ &= Ak^{\sigma} - (n+\delta)k. \end{aligned} \tag{19}$$

Maximise (19) over k:

$$\sigma Ak^{\sigma-1} = n+\delta, \tag{20}$$

that is, the marginal product of capital equals $n+\delta$. Solve (18) for s and substitute (20) for $n+\delta$:

$$\begin{aligned} s &= \frac{(n+\delta)\,k}{Ak^{\sigma}} \\ &= \frac{\big(\sigma Ak^{\sigma-1}\big)\,k}{Ak^{\sigma}} \end{aligned}$$

$$= \frac{\sigma A k^{\sigma}}{A k^{\sigma}}$$
$$= \sigma.$$

Same answer as (17).

Step — 8, sanity test at a knife edge. If $\sigma \to 0$, capital is useless, so the best saving rate must go to 0. (17) gives $s_{GR} \to 0$ and passes. The submitted answer $1/(2-\sigma)$ gives $1/2$ and fails.

Economic intuition. Saving more raises k, but each extra unit of capital adds less output and needs more replacement. The golden rule stops where one more unit of capital per worker adds exactly as much output, $\sigma A k^{\sigma-1}$, as it costs to maintain, $n+\delta$ (eq. (20)). With Cobb–Douglas, that happens when the saving rate equals capital's share.

Where the marks went. The answer wrote $c_{SS} = A(1-s)\left(\frac{sA}{n+\delta}\right)^{\frac{1}{1-\sigma}}$: it substituted k_{SS} instead of k_{SS}^{σ} (step 2 above), so the exponent lost its σ. The method (logs, derivative, solving) was right but led to $s = 1/(2-\sigma)$.

Ref: Kurlat (2020), §4.4 (golden rule); Kurlat ex. 5.8, p. 97.

What earns the points.

- c_{SS} with exponent $\sigma/(1-\sigma)$, (14).
- FOC solved to $s_{GR} = \sigma$, with the SOC or the cross-check (20).
- The interpretation: capital share.

Statement, as printed in the exam:

> **2.c (1 ponto)** Considere agora o equilíbrio competitivo de uma economia que existe por apenas dois períodos. Nesta economia há um estoque inicial de capital que se deprecia totalmente em um período. E há uma firma representativa que é maximizadora de lucros em mercados perfeitamente concorrenciais e que tem a mesma função de produção (1). O consumidor representativo escolhe o quanto consumir e trabalhar no primeiro período, mas atinge idade que o retira da força de trabalho no segundo período. Logo, suas escolhas dependem de uma riqueza inicial (em termos reais) $a_0 > 0$ e da renda real do trabalho apenas no primeiro período: $w_1 \,.\, A \,.\, L_1$. A renda não consumida no primeiro período é transferida para o segundo período através do acúmulo de riqueza, sendo r a taxa de juros real. Por fim, sua utilidade intertemporal é dada por $U = u(c_1) + \beta u(c_2)$, sendo $u(c) = ln(c_t)$. Derive e explique a restrição orçamentária intertemporal deste consumidor representativo.

Notation.

- c_1, c_2: the household's consumption in periods 1 and 2.
- $a_0 > 0$: initial real wealth. a_1: assets carried from period 1 to period 2 (saving).
- L_1: hours worked in period 1, with time endowment $\bar{L}$, normalised to $\bar{L} = 1$.
- w_1: real wage per efficiency unit of labour, so labour income is $w_1 A L_1$.
- Π_1: profit of the firm, paid to the household that owns it.
- r: real interest rate between periods 1 and 2. $\beta \in (0,1)$: discount factor (patience).
- K_1: initial capital stock; $k_1 = K_1/L_1$, $k_2 = K_2/L_1$ (capital per unit of labour).

Step — 1, who owns what. The household owns the firm. The firm uses the initial capital stock K_1, which is fully used up in production. After paying wages, the firm's profit $\Pi_1 = Y_1 - w_1 A L_1$ goes to the household.

Step — 2, period-1 budget. Resources (wealth, wages, profits) equal uses (consumption, saving):

$$c_1 + a_1 = a_0 + w_1 A L_1 + \Pi_1. \tag{21}$$

Step — 3, period-2 budget. The retired household has only its savings with interest, and it consumes everything because there is no later period:

$$c_2 = (1+r)\, a_1. \tag{22}$$

Since $c_2 \geq 0$, (22) also implies $a_1 \geq 0$: nobody lends to a household with no future income.

Step — 4, combine. Solve (22) for a_1:

$$a_1 = \frac{c_2}{1+r}.$$

Substitute this for a_1 in (21):

$$c_1 + \left(\frac{c_2}{1+r}\right) = a_0 + w_1 A L_1 + \Pi_1.$$

Drop the parentheses:

$$\boxed{c_1 + \frac{c_2}{1+r} = a_0 + w_1 A L_1 + \Pi_1} \tag{23}$$

together with the time constraint

$$0 \leq L_1 \leq 1. \tag{24}$$

Step — 5, explain. The left side of (23) is the *present value* of lifetime consumption: period-2 consumption is discounted by $1+r$. The right side is the present value of lifetime resources (wealth, wages and profits, all received in period 1). Since $u'(c) > 0$, leaving resources unspent is never optimal, so (23) holds with equality.

Economic intuition. One unit not consumed today becomes $1+r$ units tomorrow. So the price of c_2 in units of c_1 is $1/(1+r)$.

Where the marks went. The answer omitted profits Π_1 (the grader added "+Π (lucro das firmas)") and called the left side "total consumption adjusted by the interest rate" instead of its present value.

What earns the points.

- Both period constraints (21)–(22), with profits.
- The combined constraint (23).
- The present-value interpretation.

Statement, as printed in the exam:

> **2.d (1 ponto)** Monte o problema do consumidor representativo descrito em 2.c e calcule as condições de primeira ordem para **todas** as variáveis de sua escolha (incluindo a equação descrevendo a decisão intertemporal do consumo). Por fim, encontre a equação que descreve a escolha ótima de c_1 (como função apenas de parâmetros, preços e níveis de renda e riqueza), interpretando-a economicamente.

Notation.

- λ: multiplier on the budget (23) (marginal utility of a unit of wealth).
- $\nu \geq 0$: multiplier on the time constraint $L_1 \leq 1$ in (24).
- $W \equiv a_0 + w_1 A + \Pi_1$: lifetime resources when $L_1 = 1$ (defined in step 6).

Step — 1, list the choices first. The household chooses c_1, c_2 and L_1. Then (21) gives a_1. It takes w_1, r and Π_1 as given.

Step — 2, Lagrangian. Maximise U subject to (23) and (24):

$$\mathcal{L} = \ln c_1 + \beta \ln c_2 + \lambda\left[a_0 + w_1 A L_1 + \Pi_1 - c_1 - \frac{c_2}{1+r}\right] + \nu\,(1 - L_1). \tag{25}$$

Step — 3, FOC for c_1. Differentiate (25) with respect to c_1:

$$\frac{1}{c_1} = \lambda. \tag{26}$$

Step — 4, FOC for c_2. Differentiate (25) with respect to c_2:

$$\frac{\beta}{c_2} = \frac{\lambda}{1+r}. \tag{27}$$

Step — 5, FOC for L_1. Differentiate (25) with respect to L_1:

$$\lambda\, w_1 A - \nu = 0, \qquad \nu \geq 0, \qquad \nu\,(1 - L_1) = 0. \tag{28}$$

Substitute λ from (26) into (28):

$$\left(\frac{1}{c_1}\right) w_1 A - \nu = 0$$
$$\nu = \frac{w_1 A}{c_1} > 0.$$

Since $\nu > 0$, the last condition in (28) forces

$$L_1 = 1. \tag{29}$$

Work has no utility cost here (there is no leisure in U), while every hour raises income. So the household supplies its whole time endowment: a corner solution.

Step — 6, Euler equation. Substitute λ from (26) into (27):

$$\frac{\beta}{c_2} = \frac{\left(\frac{1}{c_1}\right)}{1+r}$$
$$\frac{\beta}{c_2} = \frac{1}{(1+r)\,c_1}.$$

Cross-multiply:

$$c_2 = \beta(1+r)\,c_1. \tag{30}$$

Equivalently, $\frac{1/c_1}{\beta/c_2} = 1 + r$: the marginal rate of substitution equals the relative price.

Step — 7, solve. Define lifetime resources at $L_1 = 1$:

$$W \equiv a_0 + w_1 A + \Pi_1. \tag{31}$$

Substitute (30) for c_2 and (29) for L_1 in (23):

$$c_1 + \frac{(\beta(1+r)\,c_1)}{1+r} = a_0 + w_1 A\,(1) + \Pi_1.$$

The right side is W by (31); simplify:

$$c_1 + \frac{\beta(1+r)\,c_1}{1+r} = W$$
$$c_1 + \beta c_1 = W$$
$$c_1(1+\beta) = W.$$

Hence

$$\boxed{c_1 = \frac{a_0 + w_1 A + \Pi_1}{1+\beta}}. \tag{32}$$

Substitute (32) for c_1 in (30):

$$c_2 = \beta(1+r)\left(\frac{W}{1+\beta}\right)$$

$$= \frac{\beta(1+r)}{1+\beta} W. \tag{33}$$

From (21) with $L_1 = 1$ (29) and W (31), $a_1 = W - c_1$. Substitute (32) for c_1:

$$\begin{aligned} a_1 &= W - \left(\frac{W}{1+\beta}\right) \\ &= \frac{(1+\beta)W - W}{1+\beta} \\ &= \frac{\beta}{1+\beta} W. \end{aligned} \tag{34}$$

Step — 8, interpret.

- By (32), the household consumes the fraction $1/(1+\beta)$ of lifetime resources today: its marginal propensity to consume out of wealth.
- Differentiating (32): $\partial c_1/\partial\beta = -W/(1+\beta)^2 < 0$. A more patient household consumes less today and saves more. If $\beta \to 0$, it consumes everything in period 1.
- (32) does not contain r. All resources arrive in period 1. A higher r makes saving more rewarding (substitution effect, $c_1 \downarrow$) and makes the saver richer (income effect, $c_1 \uparrow$). With log utility the two cancel exactly.

Where the marks went. The Euler equation and c_1 were right (full marks on those parts), but there was no FOC for L_1 ("Escolha de L_1??", 0/0.2), and Π_1 was again missing from c_1.

Ref: Kurlat (2020), §6.1 (two-period problem, log utility); ch. 9 for households that own firms.

What earns the points.

- Lagrangian (25) with every choice variable.
- FOCs for c_1, c_2 *and* L_1 (corner), (26)–(29).
- Euler (30), c_1 in closed form (32), interpretation of β and of r.

Statement, as printed in the exam:

> **2.e (1 ponto)** Calcule a taxa de poupança deste agente representativo (ou seja, da participação da poupança na renda do agente). Depois encontre uma expressão para k_2 como função apenas de k_1 e de parâmetros do modelo (lembrando que se trata de uma economia com mercados perfeitamente competitivos). Faça então um diagrama da expressão encontrada, colocando no eixo horizontal k_1 e no eixo vertical k_2 e mostre **graficamente** qual é o estado estacionário desta economia

Notation.

- s^*: the household's saving rate (saving over lifetime resources).
- $\phi(k_1)$: the law of motion, next period's capital as a function of this period's.

Step — 1, saving rate. Divide (34) by W:

$$\frac{a_1}{W} = \frac{\left(\frac{\beta}{1+\beta} W\right)}{W}.$$

Cancel W:

$$\boxed{s^* \equiv \frac{a_1}{W} = \frac{\beta}{1+\beta}} \tag{35}$$

It is constant and set by patience alone.

Step — 2, the firm. With $L_1 = 1$ from (29), $K_1 = k_1$. Using (1) with $A_t = A$, the firm chooses labour to maximise profit:

$$\max_{L_1} \; A k_1^{\sigma} L_1^{1-\sigma} - w_1 A L_1. \tag{36}$$

Differentiate (36) with respect to L_1:

$$(1-\sigma)Ak_1^{\sigma}L_1^{-\sigma} = w_1 A. \tag{37}$$

Substitute $L_1 = 1$ from (29) into (37) to get the wage bill:

$$\begin{aligned}(1-\sigma)Ak_1^{\sigma}\,(1)^{-\sigma} &= w_1 A \\ w_1 A &= (1-\sigma)Ak_1^{\sigma}.\end{aligned} \tag{38}$$

Step — 3, profits. Profit is output minus the wage bill, $\Pi_1 = Y_1 - w_1 A L_1$. Substitute Y_1 from (1) (with $L_1 = 1$) and $w_1 A$ from (38):

$$\begin{aligned}\Pi_1 &= \left(Ak_1^{\sigma}(1)^{1-\sigma}\right) - \left((1-\sigma)Ak_1^{\sigma}\right)(1) \\ &= Ak_1^{\sigma} - (1-\sigma)Ak_1^{\sigma} \\ &= \sigma Ak_1^{\sigma}.\end{aligned} \tag{39}$$

This is capital's share of output. Substitute (38) for $w_1 A$ and (39) for Π_1 in (31):

$$\begin{aligned}W &= a_0 + \left((1-\sigma)Ak_1^{\sigma}\right) + \left(\sigma Ak_1^{\sigma}\right) \\ &= a_0 + Ak_1^{\sigma}.\end{aligned} \tag{40}$$

Step — 4, market clearing. In a closed economy with no government, the only way to carry resources into period 2 is to install capital. So all saving becomes period-2 capital:

$$K_2 = a_1. \tag{41}$$

Substitute (34) for a_1 in (41), with $k_2 = K_2$ (one unit of labour):

$$k_2 = \left(\frac{\beta}{1+\beta}\,W\right).$$

Then substitute (40) for W:

$$k_2 = \frac{\beta}{1+\beta}(a_0 + Ak_1^{\sigma}).$$

Name this map ϕ:

$$\boxed{k_2 = \frac{\beta}{1+\beta}(a_0 + Ak_1^{\sigma}) \equiv \phi(k_1)} \tag{42}$$

Step — 5, shape of ϕ. Read off (42):

- $\phi(0) = \frac{\beta a_0}{1+\beta} > 0$: the curve starts above the origin.
- $\phi'(k) = \frac{\beta}{1+\beta}\,\sigma Ak^{\sigma-1} > 0$ and $\phi''(k) < 0$: increasing and concave.
- $\phi'(k) \to \infty$ as $k \to 0$ and $\phi'(k) \to 0$ as $k \to \infty$.

So $\phi(k) - k$ is positive at $k = 0$, eventually negative, and strictly concave. It crosses zero exactly once: one steady state, $\phi(k_{SS}) = k_{SS}$. At the crossing the curve cuts the 45° line from above, so $\phi'(k_{SS}) < 1$ and the steady state is stable.

Step — 6, steady state. For the economy to repeat, each period's household must face the same problem. Take later cohorts with no extra wealth ($a_0 = 0$; the capital they inherit already pays out through Π). Substitute $k_2 = k_1 = k$ and $a_0 = 0$ into (42):

$$\begin{aligned}(k) &= \frac{\beta}{1+\beta}\left((0) + A(k)^{\sigma}\right) \\ k &= \frac{\beta}{1+\beta}Ak^{\sigma}.\end{aligned}$$

Divide by k^σ:

$$k^{1-\sigma} = \frac{\beta A}{1+\beta}.$$

Hence

$$\boxed{k_{SS} = \left(\frac{\beta A}{1+\beta}\right)^{\frac{1}{1-\sigma}}} \tag{43}$$

The slope there: from step 5,

$$\phi'(k_{SS}) = \sigma \cdot \frac{\beta}{1+\beta} A k_{SS}^{\sigma-1}.$$

At k_{SS}, $k^{1-\sigma} = \frac{\beta A}{1+\beta}$; dividing both sides by $k_{SS}^{1-\sigma}$ gives $1 = \frac{\beta}{1+\beta} A k_{SS}^{\sigma-1}$. Substitute this bracket into $\phi'(k_{SS})$:

$$\begin{aligned}\phi'(k_{SS}) &= \sigma \cdot (1) \\ &= \sigma < 1.\end{aligned}$$

With a recurring $a_0 > 0$, k_{SS} solves $k = \phi(k)$ in (42); it has no closed form but is read off the diagram.

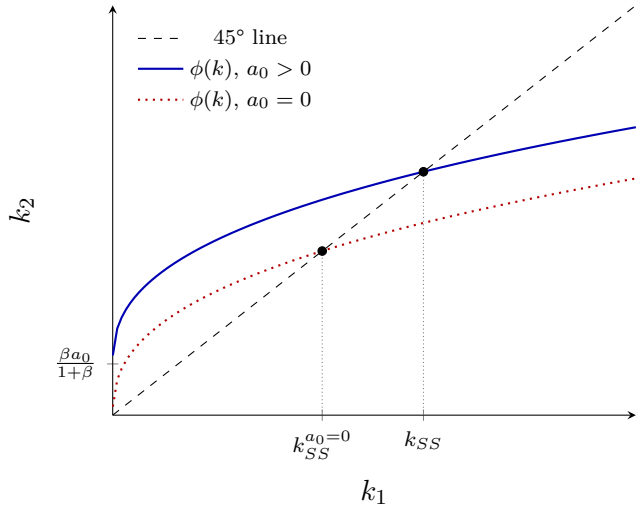


Figure 1. The law of motion (42) against the 45° line (illustrative values $A = 1$, $\sigma = 0.4$, $\beta = 0.5$, $a_0 = 0.15$). Below k_{SS} the curve lies above the 45° line, so k rises; above it, k falls.

Economic intuition. Each period the household saves a fixed fraction of what it has (35), and market clearing (41) turns that saving into next period's capital. Diminishing returns ($\sigma < 1$) make output, and so saving, grow less than proportionally with capital. Capital therefore stops growing where saving just replaces the capital that is used up.

Reading note. The statement does not say who owns K_1. In the overlapping-generations reading (the young save only their wages; the old consume the capital income), (42) becomes $k_2 = \frac{\beta}{1+\beta}(1-\sigma)Ak_1^\sigma$ and $k_{SS} = \left(\frac{\beta(1-\sigma)A}{1+\beta}\right)^{1/(1-\sigma)}$. The diagram and the argument are the same.

Where the marks went. The saving rate was not taken from the optimum ((34) already gives it). Instead the answer set $(1-s)Y = wAL$, which equates consumption with labour income and has no basis in the model. There was no market-clearing condition (41). The resulting map $k_2 = Ak_1^\sigma - w_1 A$ and the slope inequality were wrong (0/1).

Ref: Kurlat (2020), ch. 9 (equilibrium with production); Romer (2012), §2.9–2.10 (Diamond model, log utility, full depreciation), for the OLG reading.

What earns the points.

- Saving rate (35) from the household's optimum.

- Firm FOC (37), profits (39), and market clearing (41).
- $\phi(k_1)$ (42), the diagram with the 45° line, and the steady state at the crossing.

Statement, as printed in the exam:

> **2.f. (0,5 ponto)** Discuta intuitivamente a relação do modelo dos itens 2.c-2.e com o modelo de Solow dos itens 2.a-2.b.

Notation.

- k_{GR}: the golden-rule capital stock, where (20) holds.

Step — 1, same law of motion. Substitute $\delta = 1$ (capital fully used up) and $n = 0$ (constant labour) into (9):

$$\begin{aligned}(1 + (0))\, k_{t+1} &= (1 - (1))k_t + s\, Ak_t^{\sigma} \\ k_{t+1} &= s\, Ak_t^{\sigma}. \end{aligned} \tag{44}$$

With $a_0 = 0$, (42) is exactly (44) with $s = \frac{\beta}{1+\beta}$. Likewise, substitute $\delta = 1$, $n = 0$ and $s = \frac{\beta}{1+\beta}$ into (11):

$$\begin{aligned} k_{SS} &= \left(\frac{\left(\frac{\beta}{1+\beta}\right) A}{(0) + (1)} \right)^{\frac{1}{1-\sigma}} \\ &= \left(\frac{\beta A}{1+\beta} \right)^{\frac{1}{1-\sigma}}, \end{aligned}$$

which is (43).

Step — 2, the difference that matters. Solow *assumes* a constant saving rate. Here it is *chosen*, (35): it depends on patience β, and it is constant and independent of r only because utility is logarithmic. With CRRA utility of curvature $\theta \neq 1$ it would move with the interest rate. So the two-period equilibrium is a microfoundation for Solow's saving rule.

Step — 3, against the golden rule. Compare (35) with (17). Substitute $\delta = 1$ and $n = 0$ into (20):

$$\begin{aligned} \sigma A k^{\sigma-1} &= (0) + (1) \\ k^{1-\sigma} &= \sigma A \\ k_{GR} &= (\sigma A)^{\frac{1}{1-\sigma}}. \end{aligned}$$

Comparing with (43):

$$k_{SS} > k_{GR} \iff \frac{\beta}{1+\beta} > \sigma.$$

The equilibrium reaches the golden rule only by coincidence. If $\frac{\beta}{1+\beta} > \sigma$, it accumulates too much capital: saving less would raise consumption in every period (dynamic inefficiency, possible when households live finitely). Nothing in a competitive equilibrium aims at maximising steady-state consumption; households weigh today against tomorrow with their own β.

Economic intuition. Same machine (Solow's accumulation (44)), different engine (a saving rate from preferences rather than a parameter).

Where the marks went. The answer said only "Solow with $\delta = 1$ and $n = 0$" (0.1/0.5), half of step 1. It missed the endogenous saving rate and the comparison with s_{GR}.

Ref: Kurlat (2020), ex. 9.12 "Optimal vs Fixed Savings Rates", p. 185.

What earns the points.

- The mapping $\delta = 1$, $n = 0$, $s = \beta/(1 + \beta)$.
- Endogenous vs exogenous saving.
- Comparison with $s_{GR} = \sigma$.

Question 3 — The New Keynesian AS–AD model of Benigno (2015) (3.0 pt)

Statement, as printed in the exam:

> **3. (3 pontos)** Considere o modelo novo-keynesiano simplificado apresentado por Benigno (2015), que se baseia nas seguintes equações de demanda e oferta agregadas (em que variáveis com barra em cima são as de longo prazo, quando todas as firmas reajustam preços; as demais sendo de curto prazo):
>
> $$\text{(AD)} \qquad y = \bar{y} + (g - \bar{g}) - \sigma\big[i - (\bar{p} - p)\big] - \sigma ln\beta$$
>
> $$\text{(AS)} \qquad p - p^e = \kappa(y - y_n)$$
>
> em que p^e é o preço fixado pelas firmas que não podem reajustar preços e y_n é o produto natural de curto prazo, dado por: $y_n = \frac{1+\eta}{\sigma^{-1}+\eta}a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}g - \frac{1}{\sigma^{-1}+\eta}\mu$, sendo μ o parâmetro que representa o grau de markup nesta economia.

The statement's equations, with the exam's labels, for citation below:

$$y = \bar{y} + (g - \bar{g}) - \sigma\big[i - (\bar{p} - p)\big] - \sigma \ln \beta, \tag{AD}$$

$$p - p^e = \kappa\,(y - y_n), \tag{AS}$$

$$y_n = \frac{1+\eta}{\sigma^{-1}+\eta}\,a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}\,g - \frac{1}{\sigma^{-1}+\eta}\,\mu, \tag{YN}$$

$$L = \tfrac{1}{2}(y - y_e)^2 + \frac{\phi}{2\kappa}(p - p^e)^2, \tag{LF}$$

$$y_e = \frac{1+\eta}{\sigma^{-1}+\eta}\,a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}\,g. \tag{YE}$$

Notation.

- y, p: output and price level today (in logs). $\bar{y}$, $\bar{p}$: their long-run values, when all firms have reset prices.
- p^e: the price preset by the firms that cannot adjust this period.
- i: nominal interest rate, the central bank's instrument. $\beta \in (0,1)$: discount factor.
- g, $\bar{g}$: government spending today and in the long run.
- a, $\bar{a}$: productivity today and in the long run.
- μ, $\bar{\mu}$: the firms' desired markup today and in the long run.
- $\sigma > 0$: intertemporal elasticity of substitution (in Q3; it is *not* the capital share of Q2). $\eta > 0$: inverse Frisch elasticity of labour supply.
- $\kappa > 0$: slope of AS; it is larger the more firms adjust prices.
- y_n: natural output (flexible prices, with the markup). y_e: efficient output (no markup).
- $\phi \geq 0$: the weight the central bank puts on price surprises in the loss (LF).

Step — 0a, long-run output. Long-run output is (YN) evaluated at long-run values:

$$\bar{y} = \frac{1+\eta}{\sigma^{-1}+\eta}\,\bar{a} + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}\,\bar{g} - \frac{1}{\sigma^{-1}+\eta}\,\bar{\mu}. \tag{45}$$

Key fact for Q3: $\bar{\mu}$ enters through $\bar{y}$, which is in (AD). The current μ enters through y_n, which is in (AS).

Step — 0b, the two curves in the (y, p) plane. Solve (AD) for p:

$$p = \bar{p} - i - \ln\beta + \frac{\bar{y} + g - \bar{g} - y}{\sigma}. \tag{46}$$

This is a downward line with slope $-1/\sigma$. Rewrite (AS):

$$p = p^e + \kappa\,(y - y_n). \tag{47}$$

This is an upward line with slope κ through (y_n, p^e).

Step — 0c, the equilibrium. Substitute (47) for p into (AD):

$$y = \bar{y} + (g - \bar{g}) - \sigma\Big[i - \Big(\bar{p} - \big(p^e + \kappa(y - y_n)\big)\Big)\Big] - \sigma \ln \beta.$$

Open the brackets:

$$\begin{aligned} y &= \bar{y} + g - \bar{g} - \sigma i + \sigma\bar{p} - \sigma\big[p^e + \kappa(y - y_n)\big] - \sigma \ln \beta \\ y &= \bar{y} + g - \bar{g} - \sigma i + \sigma\bar{p} - \sigma p^e - \sigma\kappa\, y + \sigma\kappa\, y_n - \sigma \ln \beta. \end{aligned}$$

Move $-\sigma\kappa\, y$ to the left:

$$y + \sigma\kappa\, y = \bar{y} + g - \bar{g} - \sigma i + \sigma(\bar{p} - p^e) + \sigma\kappa\, y_n - \sigma \ln \beta.$$

Divide by $1 + \sigma\kappa$:

$$y = \frac{\bar{y} + g - \bar{g} - \sigma i + \sigma(\bar{p} - p^e) + \sigma\kappa\, y_n - \sigma \ln \beta}{1 + \sigma\kappa}. \tag{48}$$

Statement, as printed in the exam:

> **3.a (1 ponto)** Considere que há uma expectativa de aumento no longo prazo do markup. Represente graficamente, explicando economicamente o porquê, a situação de equilíbrio inicial antes desta informação e o deslocamento que define o novo equilíbrio após a nova expectativa. Não se esqueça de indicar o hiato do produto e usá-lo em sua explicação.

Step — 1, initial equilibrium E_0. Before the news, monetary policy has set i so that AD crosses AS at (y_n, p^e): output at its natural level, prices as expected, output gap $y - y_n = 0$.

Step — 2, which curve moves. Differentiate (45) with respect to $\bar{\mu}$:

$$d\bar{y} = -\frac{d\bar{\mu}}{\sigma^{-1} + \eta} < 0. \tag{49}$$

By (YN), y_n depends on the current μ, which has not moved, so $dy_n = 0$. Hence (AD) shifts left by $|d\bar{y}|$ at every p (see (46)), and **AS** (47) **does not move**. Economically: households expect lower income in the long run; to smooth consumption (Euler equation) they consume less *today* at any given real interest rate.

Step — 3, new equilibrium E_1. Differentiate (48); only $\bar{y}$, i and y_n can move. Substitute $di = 0$ and $dy_n = 0$:

$$\begin{aligned} dy &= \frac{d\bar{y} - \sigma\,(0) + \sigma\kappa\,(0)}{1 + \sigma\kappa} \\ &= \frac{d\bar{y}}{1 + \sigma\kappa}. \end{aligned} \tag{50}$$

Substitute (49) for $d\bar{y}$ in (50):

$$\begin{aligned} dy &= \frac{\left(-\dfrac{d\bar{\mu}}{\sigma^{-1} + \eta}\right)}{1 + \sigma\kappa} \\ &= -\frac{d\bar{\mu}}{(1 + \sigma\kappa)(\sigma^{-1} + \eta)} < 0. \end{aligned}$$

Differentiate (47) (p^e is preset) and substitute $dy_n = 0$:

$$\begin{aligned} dp &= \kappa\,(dy - (0)) \\ &= \kappa\, dy < 0. \end{aligned} \tag{51}$$

Step — 4, output gap. Since $dy_n = 0$, the gap moves one-for-one with output:

$$d(y - y_n) = dy < 0.$$

This is a negative output gap, a demand-driven recession, with prices below what the preset firms expected ($p < p^e$). Flexible-price firms cut prices; firms with preset prices keep p^e and produce what is demanded, so output falls.

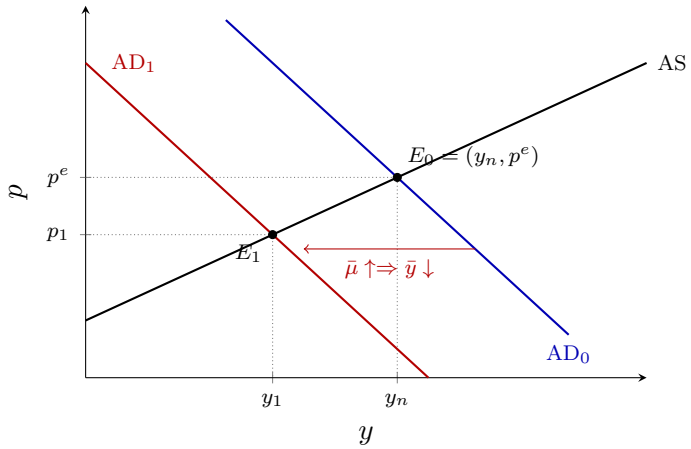


Figure 2. News of a higher long-run markup: AD shifts left, AS stays. The output gap is the horizontal distance $y_1 - y_n < 0$.

Economic intuition. The markup will raise prices relative to costs only *in the future*. What it changes today is expected income, which is a demand effect. A supply effect would need firms' costs or markups to change today.

Where the marks went. The answer shifted AS through y_n ("$\uparrow \bar{\mu}$, não $\uparrow \mu$!") and then described AD with $p \uparrow, y \downarrow$, a contradictory sign story (0.2/1).

Ref: Benigno (2015), §6.3 (news about the future, Fig. 8, p. 513) and §7 (markup shocks, Fig. 9).

What earns the points.

- AD shifts left because $\bar{y}$ falls (49); AS fixed because y_n is fixed.
- Graph with E_0 and E_1.
- Negative gap (50) and the mechanism (consumption smoothing, sticky prices).

Statement, as printed in the exam:

> **3.b (1 ponto)** Com base no item anterior, explique intuitivamente o que uma autoridade monetária comprometida com a estabilidade de preços deveria fazer após a expectativa de aumento no markup, e qual o custo econômico de assim proceder.

Step — 1, the target. Price stability means $p = p^e$. By (47), that requires $y = y_n$, i.e. back to E_0. Monetary policy cannot move AS; it moves AD through i.

Step — 2, the size of the move. Differentiate (48) and substitute $dy = 0$ (the target) and $dy_n = 0$:

$$\begin{aligned}(0) &= \frac{d\bar{y} - \sigma\, di + \sigma\kappa\,(0)}{1 + \sigma\kappa} \\ 0 &= d\bar{y} - \sigma\, di \\ \sigma\, di &= d\bar{y}.\end{aligned}$$

Hence

$$di = \frac{d\bar{y}}{\sigma}. \tag{52}$$

Substitute (49) for $d\bar{y}$ in (52):

$$\begin{aligned} di &= \frac{\left(-\dfrac{d\bar{\mu}}{\sigma^{-1}+\eta}\right)}{\sigma} \\ &= -\frac{d\bar{\mu}}{\sigma(\sigma^{-1}+\eta)} \\ &= -\frac{d\bar{\mu}}{\sigma\sigma^{-1}+\sigma\eta} \\ &= -\frac{d\bar{\mu}}{1+\sigma\eta} < 0. \end{aligned} \tag{53}$$

The central bank **cuts** i. The cut lowers the real rate, raises current consumption, and moves AD_1 back to AD_0 in Figure 2. Its size equals the fall in the natural rate of interest.

Step — 3, the cost.

- In terms of the output gap the cost is *zero*. A pure demand shock has no trade-off: stabilising prices also closes $y - y_n$ (what the literature calls the divine coincidence).
- What policy *cannot* fix: output returns only to y_n, which is below the efficient y_e (compare (YN) with (YE): the markup term), and $\bar{y}$ is permanently lower by (49). Those are real effects, beyond monetary policy.
- If i is already near zero, the cut in (53) may be infeasible (zero lower bound, §9). The recession then persists unless policy can promise a higher $\bar{p}$ or use fiscal policy.

Economic intuition. Raising i, as in the answer, would push AD further left by (48): a deeper recession and a larger fall in prices, the opposite of price stability.

Where the marks went. The answer said to *raise* the rate "to contain the rise in prices"; prices are falling here. The cost was not discussed (0.1/1).

Ref: Benigno (2015), §6 (natural rate and policy), §9 (zero lower bound), §11 (no trade-off for demand shocks).

What earns the points.

- Cut i, (53), with the reason (AD back to E_0).
- No gap cost (divine coincidence).
- What remains: $y_n < y_e$, lower $\bar{y}$, the ZLB caveat.

Statement, as printed in the exam:

> **3.c (1 ponto)** Suponha que a autoridade monetária representa o conjunto de preferências sociais e seus objetivos em termos da seguinte função de perda quadrática e do parâmetro $\phi \geq 0$:
>
> $$\text{(LF)} \qquad L = \frac{1}{2}(y - y_e)^2 + \frac{\phi}{2\kappa}(p - p^e)^2$$
>
> onde $y_e = \frac{1+\eta}{\sigma^{-1}+\eta}a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}g$ é o nível eficiente de produto. Derive a regra ótima de política monetária (IT) e mostre graficamente qual é a política ótima após a expectativa de aumento do markup no longo prazo. Discuta como sua resposta neste item depende do valor de ϕ.

Notation.

- y^*, p^*, i^*: the optimal output, price level and interest rate.

Step — 1, the constraint. The central bank sets i, which moves only AD. So it can pick any point on AS, and (AS) is its constraint:

$$p - p^e = \kappa\,(y - y_n). \tag{54}$$

Step — 2, substitute the constraint. Substitute (54) for $p - p^e$ in (LF):

$$L = \tfrac{1}{2}(y - y_e)^2 + \frac{\phi}{2\kappa}\left(\kappa(y - y_n)\right)^2$$

$$= \tfrac{1}{2}(y - y_e)^2 + \frac{\phi}{2\kappa}\,\kappa^2(y - y_n)^2$$
$$= \tfrac{1}{2}(y - y_e)^2 + \frac{\phi\kappa}{2}(y - y_n)^2. \tag{55}$$

This is the loss as a function of y alone, $L(y)$.

Step — 3, first-order condition. Differentiate (55) and set it to zero:

$$(y - y_e) + \phi\kappa\,(y - y_n) = 0. \tag{56}$$

The second derivative of (55) is $1 + \phi\kappa > 0$, so (56) gives a minimum.

Step — 4, the targeting rule. Group (56) as $(y - y_e) + \phi\big(\kappa(y - y_n)\big) = 0$. By (54), the bracket equals $p - p^e$. Substitute it:

$$(y - y_e) + \phi(p - p^e) = 0.$$

That is the targeting rule:

$$\boxed{(y - y_e) + \phi\,(p - p^e) = 0}. \tag{57}$$

Step — 5, the optimal point. Solve (56) for y:

$$y(1 + \phi\kappa) = y_e + \phi\kappa\, y_n$$
$$y^* = \frac{y_e + \phi\kappa\, y_n}{1 + \phi\kappa}. \tag{58}$$

Subtract y_n from (58):

$$y^* - y_n = \frac{y_e + \phi\kappa\, y_n}{1 + \phi\kappa} - y_n$$
$$= \frac{y_e + \phi\kappa\, y_n - (1 + \phi\kappa)\, y_n}{1 + \phi\kappa}$$
$$= \frac{y_e - y_n}{1 + \phi\kappa}.$$

Substitute this for $y - y_n$ in (54):

$$p^* - p^e = \kappa\left(\frac{y_e - y_n}{1 + \phi\kappa}\right)$$
$$p^* = p^e + \frac{\kappa\,(y_e - y_n)}{1 + \phi\kappa}. \tag{59}$$

Step — 6, the instrument. Substitute (y^*, p^*) for (y, p) in (AD):

$$(y^*) = \bar{y} + (g - \bar{g}) - \sigma\big[i - (\bar{p} - (p^*))\big] - \sigma \ln\beta.$$

Solve for i:

$$\sigma i = \bar{y} + g - \bar{g} - y^* + \sigma(\bar{p} - p^*) - \sigma \ln\beta$$
$$i^* = \frac{\bar{y} + g - \bar{g} - y^*}{\sigma} + \bar{p} - p^* - \ln\beta. \tag{60}$$

Step — 7, the news. $\bar{\mu}$ appears in neither (YE) nor (YN), so by (58) and (59) y^* and p^* do not change. Only $\bar{y}$ changes. Differentiate (60) and substitute $dy^* = dp^* = 0$:

$$di^* = \frac{d\bar{y} - (0)}{\sigma} - (0)$$
$$= \frac{d\bar{y}}{\sigma}.$$

Substitute (49) for $d\bar{y}$:

$$di^* = \frac{\left(-\dfrac{d\bar{\mu}}{\sigma^{-1}+\eta}\right)}{\sigma} = -\frac{d\bar{\mu}}{1+\sigma\eta} < 0. \tag{61}$$

ϕ enters (60) only through y^* and p^*, which did not move, so (61) does not depend on ϕ. The central bank fully offsets the AD shift for *every* weight: it is the same cut as (53).

Step — 8, the role of ϕ. ϕ decides *where* on AS the economy sits, not how policy reacts to this shock. With $\mu > 0$, (YN) and (YE) give $y_e > y_n$. Then, from (58)–(59):

- $\phi \to \infty$ (only prices matter): $y^* = y_n$, $p^* = p^e$, strict price stability.
- $\phi = 0$ (only output matters): $y^* = y_e$, $p^* > p^e$.
- $0 < \phi < \infty$: $y_n < y^* < y_e$; the bank accepts some price surprise to bring output closer to its efficient level.

ϕ *would* matter for a *current* markup shock, which moves y_n in (YN) but not y_e in (YE), and so creates a trade-off.

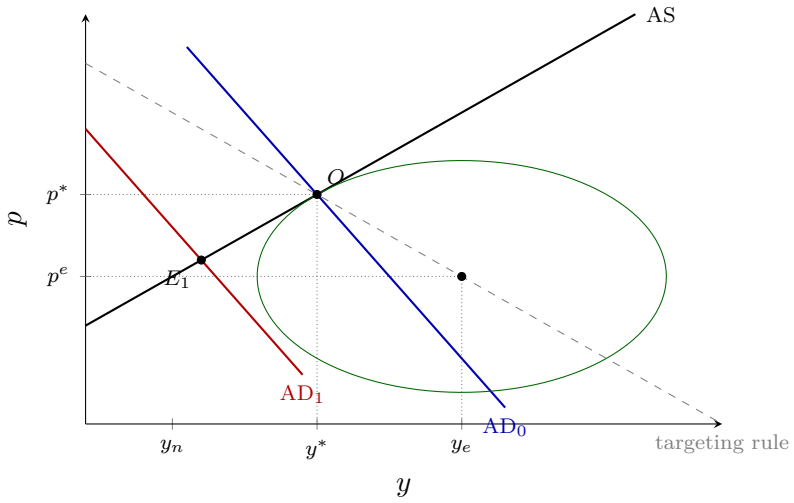


Figure 3. Optimal policy (illustrative values $\kappa = 0.5$, $\sigma = 1$, $\phi = 2$, $y_n = 0$, $y_e = 1$, $p^e = 0$). The loss ellipse centred at (y_e, p^e) is tangent to AS at O, where the targeting rule (57) crosses AS. The news moves AD_0 to AD_1 (point E_1 if i is unchanged); the optimal cut (61) brings it back to AD_0, through O.

Economic intuition. The loss function says what the bank cares about; AS says what it can get. The news changes neither, so the best attainable point does not change. Only the interest rate needed to reach it changes.

Where the marks went. The answer did not set $dL/dy = 0$ along AS (step 3). It substituted AS into AD, differentiated with respect to i, and then used a y_e containing μ, concluding that the response depends on whether $\phi > \sigma$. For this shock the response (61) does not depend on ϕ at all.

Ref: Benigno (2015), §11, eqs. (33)–(35), Fig. 14, pp. 521–523; Lista 7 Q4.

What earns the points.

- Loss written on AS, (55), and $dL/dy = 0$ with SOC.
- Targeting rule (57), y^*, p^*, i^*.
- Graph (ellipse tangent to AS, AD shifted and restored).
- (61) independent of ϕ, and what ϕ does decide.